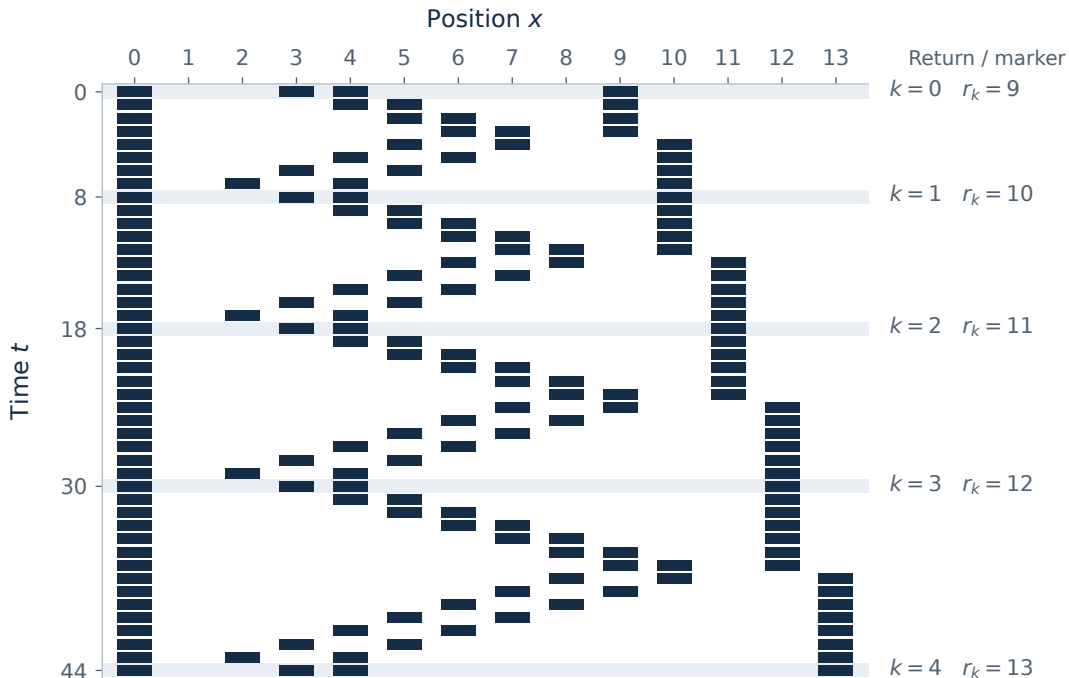


Binary four-particle shuttle

$d = 9$; initial support $\{0, 3, 4, 9\}$



$$t_k = k^2 + (2d - 11)k = k^2 + 7k, \quad r_k = d + k$$

Return intervals: 8, 10, 12, 14 steps. The right marker advances one site per cycle.